

# Proof that $\sqrt{7}$ is Irrational

## Problem

Prove that  $\sqrt{7}$  is irrational.

## Solution

The number  $\sqrt{7}$  is irrational.

**Proof. Step 1: Every rational number has a representation in lowest terms.** If  $r \in \mathbb{Q}$ , then by definition there exist integers  $a, b$  with  $b \neq 0$  such that  $r = a/b$ . Let  $d = \gcd(a, b) > 0$ . Write  $a = dp$  and  $b = dq$  for integers  $p, q$ ; then  $q \neq 0$  and

$$r = \frac{a}{b} = \frac{dp}{dq} = \frac{p}{q}.$$

We verify  $\gcd(p, q) = 1$ : if a positive integer  $c$  divides both  $p$  and  $q$ , then  $dc$  divides both  $a = dp$  and  $b = dq$ , so  $dc \leq d = \gcd(a, b)$ , giving  $c \leq 1$ , hence  $c = 1$ .

**Step 2: A key lemma on divisibility by 7.** For every integer  $n$ , if  $7 \mid n^2$  then  $7 \mid n$ .

*Proof of Lemma ??.* We use the fact that congruence is compatible with multiplication: if  $n \equiv r \pmod{7}$ , then

$$n^2 - r^2 = (n - r)(n + r).$$

Since  $7 \mid (n - r)$ , it follows that  $7 \mid n^2 - r^2$ , i.e.  $n^2 \equiv r^2 \pmod{7}$ .

Every integer  $n$  satisfies  $n \equiv r \pmod{7}$  for exactly one  $r \in \{0, 1, 2, 3, 4, 5, 6\}$ . By the compatibility just established,  $n^2 \equiv r^2 \pmod{7}$ . Computing:

$$0^2 \equiv 0, \quad 1^2 \equiv 1, \quad 2^2 \equiv 4, \quad 3^2 \equiv 2, \quad 4^2 \equiv 2, \quad 5^2 \equiv 4, \quad 6^2 \equiv 1 \pmod{7}.$$

The only residue  $r$  for which  $r^2 \equiv 0 \pmod{7}$  is  $r = 0$ . Hence  $n^2 \equiv 0 \pmod{7}$  implies  $n \equiv 0 \pmod{7}$ , i.e.  $7 \mid n$ .  $\square$

**Step 3: The main argument.** Suppose for contradiction that  $\sqrt{7}$  is rational. By Step 1, write  $\sqrt{7} = p/q$  with  $p, q \in \mathbb{Z}$ ,  $q \neq 0$ , and  $\gcd(p, q) = 1$ . Squaring and multiplying by  $q^2$  gives

$$p^2 = 7q^2.$$

Hence  $7 \mid p^2$ , and Lemma ?? gives  $7 \mid p$ . Write  $p = 7k$  for some integer  $k$ . Substituting:

$$(7k)^2 = 7q^2 \implies 49k^2 = 7q^2 \implies 7k^2 = q^2.$$

Hence  $7 \mid q^2$ , and Lemma ?? gives  $7 \mid q$ .

Thus 7 divides both  $p$  and  $q$ , contradicting  $\gcd(p, q) = 1$ . Therefore  $\sqrt{7}$  is irrational.  $\square$